Exercise 3: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 3

From Fundamentals to AI Apocalypse · personal study guide

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Exercise 3.1: Mathematical induction

Induction separates an infinite claim into a base case and a step showing each case implies the next.

Connected topic: Mathematical induction (§2.5) gives the two-part proof structure used in both claims.

1a: Prove the square-sum inequality

Try first: write the base case, then add the new final term $(n + 1)^2$ to the induction hypothesis.

Answers: see "Exercise 3.1: Mathematical induction" under Answers at the end.

1b: Prove 4 divides $5^n - 1$

Try first: check $n = 1$ and rewrite the next power using $5 \cdot 5^n$.

Answers: see "Exercise 3.1: Mathematical induction" under Answers at the end.

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Exercise 3.2: Maps

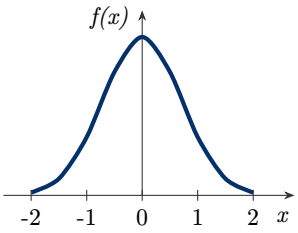
Injectivity asks whether different inputs can share an output; surjectivity asks whether every target value is reached. A bijection has both properties.

Connected topics: Types of mappings (§3.2), inverse functions (§3.3), and composition (§3.4).

2a: Classify $f(x) = e^{-x^2}$ and $g(x,y) = (4y, x-1)$

Try first: find the range and look for two different inputs with the same output; for g, solve its two coordinate equations.

Answers: see "Exercise 3.2: Maps" under Answers at the end.



$f(x) = e^{-x^2}$. The even graph shows why f(x) = f(−x); its values stay positive and at most 1.

2b(i): Decide whether f ∘ g can be formed

Try first: compare the output type of the inner map g with the input type required by the outer map f.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

2b(ii): Find f ∘ f

Try first: substitute f(x) into the outer f and simplify the square in the exponent.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

2b(iii): Find g ∘ g

Try first: substitute (4y, x−1) for the input pair of the outer g.

Answers: see "Exercise 3.2: Maps" under Answers at the end.

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Exercise 3.3: Properties of the binomial coefficient

All three identities follow from the factorial definition; the last is Pascal's rule.

Connected topic: Binomial coefficients (§2.4) introduces factorials and Pascal's identity.

3i: Prove C(n,n) = C(n,0) = 1

Try first: substitute k = n and k = 0 into the factorial definition, using 0! = 1.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

3ii: Prove C(n,k) = C(n,n−k)

Try first: write both sides using factorials and compare the denominator factors.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

3iii: Prove Pascal's identity

Try first: put the two left-hand fractions over the common denominator (k+1)!(n−k)!.

Answers: see "Exercise 3.3: Properties of the binomial coefficient" under Answers at the end.

Exercise 3.4: Elementary functions

Substituting x^4 reduces the degree-eight polynomial to a quadratic. The number of real roots of the parameterized quadratic is controlled by its discriminant.

Connected topics: Elementary functions (§3.5) and the quadratic formula and inequalities (§2.3).

4a: Find and factor the real roots of $f(x) = x^8 - 3x^4 - 4$

Try first: set $t = x^4$. Remember t cannot be negative, even though the resulting quadratic has two algebraic roots.

Answers: see "Exercise 3.4: Elementary functions" under Answers at the end.

4b: Choose c so $g_c(x) = 3x^2 + 2x + c$ has exactly one real root

Try first: set the discriminant to zero, then find the repeated root and write the factorization.

Answers: see "Exercise 3.4: Elementary functions" under Answers at the end.

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Self-Study 3.5: Mathematical induction

The finite geometric-sum formula is proved by adding exactly one new power and simplifying over the fixed denominator.

Connected topic: Mathematical induction (§2.5) supplies the base-step method.

5: Prove the finite geometric-sum formula

Try first: use $n = 1$ as the base case; in the step, add x^{n+1} to the assumed sum.

Answers: see "Self-Study 3.5: Mathematical induction" under Answers at the end.

Self-Study 3.6: Power sums

Each identity is proved independently: the induction step adds the new term $(n+1)$, $(n+1)^2$, or $(n+1)^3$ and factors the result.

Connected topic: Induction (§2.5) supports each proof; the sum notation is reviewed in §2.4.

6a: Prove $\Sigma_{k=1}^{n}\, k = n(n+1)/2$

Try first: check $n = 1$, then add $n+1$ to the formula at n.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

6b: Prove $\Sigma_{k=1}^{n}\, k^2 = n(n+1)(2n+1)/6$

Try first: after adding $(n+1)^2$, factor out $n+1$ and compare with the target at $n+1$.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

6c: Prove $\Sigma_{k=1}^{n}\, k^3 = (n(n+1)/2)^2$

Try first: add $(n+1)^3$ to the squared induction hypothesis, then factor $(n+1)^2$.

Answers: see "Self-Study 3.6: Power sums" under Answers at the end.

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Self-Study 3.7: Maps

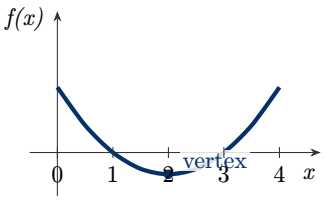
Vertex form gives the range and symmetry of the quadratic. For compositions, the inner output must lie in the outer function's domain.

Connected topics: Types of mappings (§3.2), inverse functions (§3.3), and composition (§3.4).

7a: Classify f, g and h

Try first: write f in vertex form, determine the actual range of g, and solve h(x,y,z)=(u,v) for arbitrary u,v.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.



$f(x)=(x-2)^2-1$. The symmetry around x=2 explains why f is not injective on all of $\mathbb{R}$; restricting to one side of the vertex makes it invertible.

7b(i): Form f ∘ s

Try first: substitute sin x into f, then find the range by examining the resulting quadratic for t=sin x∈[−1,1].

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(ii): Form g ∘ f

Try first: require f(x)≤1 before substituting it under the square root.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(iii): Form g ∘ s

Try first: check whether every value of sin x lies in g's domain, then use the extreme sine values for the range.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

7b(iv): Form g ∘ g

Try first: require the output of the inner g to be at most 1, not just that the original x is at most 1.

Answers: see "Self-Study 3.7: Maps" under Answers at the end.

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Self-Study 3.8: Finite decimals and computer arithmetic

Finite decimals form a set closed under addition and multiplication, but they do not include every rational. Truncation gives controlled approximations; floating-point arithmetic can change familiar algebraic laws.

Connected topics: Number systems (§2.1) and real-number arithmetic (§2.2).

8a: Prove D is closed under addition and multiplication

Try first: put two denominators over a common power of 10 for the sum; multiply numerators and denominators for the product.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8b: Prove $1/3 \notin D$

Try first: assume $1/3 = m/10^n$ and examine the resulting equality modulo 3.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8c: Prove the truncation bounds

Try first: apply the defining inequality for $\text{floor}(10^n x)$, then divide by the positive number 10^n.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

8d: Compare the two floating-point sums

Try first: evaluate the parenthesized operations in order and remember that a double cannot represent every integer near 10^{16}.

Answers: see "Self-Study 3.8: Finite decimals and computer arithmetic" under Answers at the end.

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Self-Study 3.9: Fibonacci numbers in Pascal's triangle

The diagonal sums in Pascal's triangle satisfy the same two-step recurrence as the Fibonacci sequence.

Connected topics: Binomial coefficients and Pascal's identity (§2.4), and induction (§2.5).

9a: Compute F_1 through F_{10}

Try first: start from $F_1=F_2=1$ and add the previous two values at each step.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

9b: List Pascal's rows 0–7 and compute D_0–D_7

Try first: for each n, add $C(n-k,k)$ across k; use the stated zero convention when $k>n-k$.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

9c: Prove $D_n = F_{n+1}$ by induction

Try first: derive $D_{n+2}=D_{n+1}+D_n$ from Pascal's identity, then use two initial values.

Answers: see "Self-Study 3.9: Fibonacci numbers in Pascal's triangle" under Answers at the end.

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Answers

Exercise 3.1: Mathematical induction

1a · Induction on n

1. **Claim:** $S_n = \sum_{k=1}^{n}(k-1)^2 < n^3/3$.
2. **Base case $n = 1$:** $S_1 = (1-1)^2 = 0 < 1/3$, so the claim starts true.
3. **Induction hypothesis:** assume $S_n < n^3/3$ for some $n \in \mathbb{N}$.
4. The next sum adds the term $k = n + 1$, which equals n^2:

$$S_{n+1} = S_n + n^2 < n^3/3 + n^2$$

5. Compare the upper bound with the target:

$$(n+1)^3/3 = n^3/3 + n^2 + n + 1/3 > n^3/3 + n^2$$

6. Therefore $S_{n+1} < (n+1)^3/3$. The base and step prove the claim for every $n \in \mathbb{N}$.

Exercise 3.1: Mathematical induction

1b · Preserve divisibility in the induction step

1. **Base case $n = 1$:** $5^1 - 1 = 4$, divisible by 4.
2. **Induction hypothesis:** suppose $5^n - 1 = 4q$ for some integer q.
3. Then the next expression is:

$$5^{n+1} - 1 = 5 \cdot 5^n - 1 = 5(5^n - 1) + 4$$

4. Substitute the hypothesis: $5(4q) + 4 = 4(5q + 1)$, a multiple of 4. Hence 4 divides $5^n - 1$ for every positive integer n.

Made

Exercise 3.2: Maps

2a · Range and one-to-one tests

1. For f, $x^2 \geq 0$, so $-x^2 \leq 0$ and $0 < e^{-x^2} \leq 1$. The maximum 1 occurs at $x = 0$; the value 0 is approached as $|x|$ grows but is never reached. Thus range(f) = (0,1], so f is not onto $\mathbb{R}$.
2. Also $f(1) = f(-1) = e^{-1}$ with distinct inputs, so f is not injective. Therefore it is not bijective as stated.
3. Restrict its domain to $[0,\infty)$ and its range/codomain to (0,1]. On this domain f decreases strictly from 1 toward 0, so the restricted map is bijective. Solve $y = e^{-x^2}$ with $x \geq 0$:

$$\ln y = -x^2 \Rightarrow x = \sqrt{-\ln y}, \quad 0 < y \leq 1$$

   Its inverse is $f^{-1}(y) = \sqrt{-\ln y}$.
4. For g, if $g(x_1,y_1) = g(x_2,y_2)$, coordinate equality gives $4y_1 = 4y_2$ and $x_1 - 1 = x_2 - 1$, hence both input coordinates agree. So g is injective.
5. Given any target $(u,v) \in \mathbb{R}^2$, choose $x = v + 1$ and $y = u/4$. Then $g(x,y) = (u,v)$, so g is surjective as well. It is bijective, with:

$$g^{-1}(u, v) = (v + 1, u/4)$$

Exercise 3.2: Maps

2b(i) · Match codomain to domain

1. g takes a vector in $\mathbb{R}^2$ and returns a vector in $\mathbb{R}^2$. But f expects a real number, an element of $\mathbb{R}$.
2. The codomain $\mathbb{R}^2$ of g is not contained in the domain $\mathbb{R}$ of f. Thus $f \circ g$ is **not defined** with the stated maps.

Exercise 3.2: Maps

2b(ii) · Compose the real-valued map with itself

1. The output of f lies in $\mathbb{R}$, so it is a valid input for f. The composition has domain $\mathbb{R}$ and mapping rule:

$$(f \circ f)(x) = e^{-(e^{-x^2})^2} = e^{-e^{-2x^2}}$$

2. The inner value $t = f(x)$ ranges over (0,1]. Applying f to such a t gives e^{-t^2}, which ranges from e^{-1} (at $t = 1$) up toward, but never reaches, 1 (as t approaches 0).
3. Therefore range$(f \circ f) = [e^{-1},1)$.

Exercise 3.2: Maps

2b(iii) · Compose the plane map

1. Both copies of g map $\mathbb{R}^2$ to $\mathbb{R}^2$, so the composition is defined on all of $\mathbb{R}^2$.
2. Apply the rule twice:

$$(g \circ g)(x, y) = g(4y, x - 1) = (4(x - 1), 4y - 1)$$

3. The result is an invertible affine map, so its range is $\mathbb{R}^2$. Explicitly, any target (u,v) comes from $x = u/4 + 1$ and $y = (v+1)/4$.

Exercise 3.3: Properties of the binomial coefficient

3i · The two endpoint choices

1. By definition $C(n,k) = n!/[k!(n-k)!]$. At $k = n$:

$$C(n, n) = n!/(n! \cdot 0!) = 1$$

2. At $k = 0$:

$$C(n, 0) = n!/(0! \cdot n!) = 1$$

3. Each coefficient counts the unique way to choose either all n objects or none.

Exercise 3.3: Properties of the binomial coefficient

3ii · Symmetry under exchanging chosen and omitted objects

1. For $0 \leq k \leq n$:

$$C(n, n - k) = n!/((n - k)! \cdot (n - (n - k))!) = n!/((n - k)! \cdot k!)$$

2. Factorial multiplication is commutative, so this denominator equals $k!(n-k)!$. That is $C(n,k)$, proving the identity.
3. Combinatorially, choosing k objects is equivalent to choosing the $n-k$ objects left behind.

Exercise 3.3: Properties of the binomial coefficient

3iii · Add adjacent binomial coefficients

1. Using the factorial definition, the left side is:

$$C(n,k) + C(n,k+1) = n!/[k!(n-k)!] + n!/[(k+1)!(n-k-1)!]$$

2. Use the common denominator $(k+1)!(n-k)!$:

$$= ((k+1)n! + (n-k)n!)/((k+1)!(n-k)!) = (n+1)!/((k+1)!(n-k)!)$$

3. The final fraction is $C(n+1,k+1)$. The stated range $0 \le k \le n-1$ ensures all factorials above have non-negative integer arguments.

Exercise 3.4: Elementary functions

4a · Reduce to a quadratic in x^4

1. Let $t = x^4 \ge 0$. Then $f(x) = t^2 - 3t - 4$ factors as $(t-4)(t+1)$.
2. Setting $f(x) = 0$ gives $t = 4$ or $t = -1$. The second value is impossible because $t = x^4 \ge 0$.
3. Thus $x^4 = 4$. Equivalently $x^2 = 2$, so the real roots are $x = \sqrt{2}$ and $x = -\sqrt{2}$.
4. Factor fully over $\mathbb{R}$:

$$x^8 - 3x^4 - 4 = (x-\sqrt{2})(x+\sqrt{2})(x^2+2)(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$$

5. The remaining quadratics have no real roots: x^2+2 is always positive, and each $x^2 \pm \sqrt{2}x + 1$ has discriminant $2-4 = -2$. So the two displayed linear factors are all the real linear factors.

Exercise 3.4: Elementary functions

4b · One distinct root means a repeated root

1. For ax^2+bx+c the discriminant is b^2-4ac. Here $a = 3$ and $b = 2$, so $\Delta = 4-12c$.
2. A real quadratic with non-zero leading coefficient has exactly one distinct real root precisely when $\Delta = 0$. Therefore $4-12c = 0$, giving $c = 1/3$.
3. The repeated root is $-b/(2a) = -2/6 = -1/3$. The corresponding factorization is:

$$g_{1/3}(x) = 3x^2 + 2x + 1/3 = 3(x + 1/3)^2$$

4. **Check:** expanding gives $3x^2+2x+1/3$, and its only distinct root is $-1/3$ (with multiplicity two).

Self-Study 3.5: Mathematical induction

5 · Induction for every $x \neq 1$

1. Fix any real $x \neq 1$. The claim is $\Sigma_{k=0}{}^{n} x^k = (1-x^{n+1})/(1-x)$.
2. **Base n = 1:** $1+x = (1-x^2)/(1-x)$, since $1-x^2 = (1-x)(1+x)$.
3. **Induction hypothesis:** suppose the formula holds for n. Add the next term:

$$\sum_{k=0}^{n+1} x^k \;=\; (1-x^{n+1})/(1-x) \;+\; x^{n+1}$$

4. Put the terms over one denominator and simplify:

$$= (1-x^{n+1} + x^{n+1}(1-x))/(1-x) \;=\; (1-x^{n+2})/(1-x)$$

5. This is the desired formula at n+1. Thus it holds for every $n \in \mathbb{N}$ and each fixed $x \in \mathbb{R}\backslash\{1\}$.

Self-Study 3.6: Power sums

6a · Sum of the first n integers

1. **Base:** $1 = 1\cdot 2/2$.
2. **Assume** $\Sigma_{k=1}{}^{n} k = n(n+1)/2$. Then:

$$\sum_{k=1}^{n+1} k \;=\; n(n+1)/2 \;+\; (n+1) \;=\; (n+1)(n+2)/2$$

3. This is the same formula with n replaced by n+1, proving it for all positive n.

Self-Study 3.6: Power sums

6b · Sum of the first n squares

1. **Base:** $1^2 = 1\cdot 2\cdot 3/6$.
2. **Assume** $S_n = n(n+1)(2n+1)/6$. Add the next square:

$$S_{n+1} \;=\; n(n+1)(2n+1)/6 \;+\; (n+1)^2$$

3. Factor n+1 and combine:

$$= (n+1)[n(2n+1) + 6(n+1)]/6 \;=\; (n+1)(2n^2+7n+6)/6$$

4. Since $2n^2+7n+6 = (n+2)(2n+3)$, this equals $(n+1)(n+2)(2n+3)/6$, the formula at n+1.

Self-Study 3.6: Power sums

6c · Sum of the first n cubes

1. **Base:** $1^3 = (1{\cdot}2/2)^2$.
2. **Assume** $S_n = [n(n+1)/2]^2$. The next sum is:

$$S_{n+1} = [n(n+1)/2]^2 + (n+1)^3$$

3. Factor $(n+1)^2$:

$$= (n+1)^2[n^2/4 + n + 1] = (n+1)^2(n+2)^2/4 = [(n+1)(n+2)/2]^2$$

4. This is the claimed formula for n+1, completing the induction.

Self-Study 3.7: Maps

7a · Injective, surjective and bijective

1. Complete the square for f:

$$f(x) = x^2 - 4x + 3 = (x-2)^2 - 1$$

   Its minimum is -1 at $x = 2$, so range(f) = $[-1,\infty)$. It is not injective because $f(1)=f(3)=0$, and it is not onto $\mathbb{R}$ because it never takes values below -1.

2. Restrict f to $[2,\infty)$ and set its codomain to $[-1,\infty)$. It is then strictly increasing and onto that range. Solving $y=(x-2)^2-1$ with $x\geq 2$ gives the inverse:

$$f^{-1}(y) = 2 + \sqrt{y+1}, \quad y \geq -1$$

3. For $g(x)=\sqrt{1-x}$, the stated domain $x\leq 1$ makes the radicand non-negative. The function is strictly decreasing, so it is injective. Its range is $[0,\infty)$, not all $\mathbb{R}$, so it is not surjective onto $\mathbb{R}$.
4. Keep the domain $(-\infty,1]$ but change the codomain to $[0,\infty)$. This is a bijection. Solving $y=\sqrt{1-x}$ gives $x=1-y^2$, so $g^{-1}(y)=1-y^2$ for $y\geq 0$.
5. For $h(x,y,z)=(x+y,y+z)$, every $(u,v)\in\mathbb{R}^2$ is reached by choosing $(x,y,z)=(u,0,v)$, hence h is surjective.
6. It is not injective: adding $(-t,t,-t)$ to any input leaves both output coordinates unchanged. More explicitly, all preimages of (u,v) have the form $(u-t,t,v-t)$, $t\in\mathbb{R}$. Thus h is not bijective.

Self-Study 3.7: Maps

7b(i) · Compose f with sine

1. s maps $\mathbb{R}$ to $[-1,1]$, which lies in the domain of f. So $f \circ s$ is defined for every real x:

$$(f \circ s)(x) = \sin^2 x - 4\sin x + 3$$

2. Let $t=\sin x$. The polynomial t^2-4t+3 has derivative $2t-4<0$ throughout $[-1,1]$, so it decreases across that interval.
3. At $t=1$ its value is 0; at $t=-1$ its value is 8. Since sine attains every value between -1 and 1, the range is $\mathbf{[0,8]}$.

Self-Study 3.7: Maps

7b(ii) · Restrict inputs to g's domain

1. g accepts inputs at most 1. Since $f(x)=(x-2)^2-1$, the composition domain satisfies:

$$(x-2)^2 - 1 \le 1 \Leftrightarrow (x-2)^2 \le 2 \Leftrightarrow 2-\sqrt{2} \le x \le 2+\sqrt{2}$$

2. On that interval:

$$(g \circ f)(x) = \sqrt{1-f(x)} = \sqrt{2-(x-2)^2}$$

3. The radicand ranges from 0 at the two endpoints to 2 at $x=2$. Therefore the range is $[0,\sqrt{2}]$.

Self-Study 3.7: Maps

7b(iii) · Sine always stays in the allowed domain

1. For every real x, $\sin x \le 1$. Thus $s(x)$ belongs to the domain $(-\infty,1]$ of g and $g \circ s$ has domain $\mathbb{R}$.
2. The rule is:

$$(g \circ s)(x) = \sqrt{1-\sin x}$$

3. Since $-1 \le \sin x \le 1$, the radicand ranges from 0 to 2, and both endpoints occur. The range is $[0,\sqrt{2}]$.

Self-Study 3.7: Maps

7b(iv) · Restrict the inner output

1. The first g requires $x \leq 1$. Its output $\sqrt{1-x}$ must also be in g's domain, so $\sqrt{1-x} \leq 1$. Squaring non-negative sides gives $x \geq 0$.
2. The composite domain is therefore [0,1], and its rule is:

$$(g \circ g)(x) = \sqrt{1-\sqrt{1-x}}$$

3. As x increases from 0 to 1, the inner square root decreases from 1 to 0, so the outer expression increases continuously from 0 to 1. The range is [0,1].

Self-Study 3.8: Finite decimals and computer arithmetic

8a · Closure of finite decimals

1. Take $m/10^n$ and $p/10^r$ in D, where $m,p \in \mathbb{Z}$ and $n,r \in \mathbb{N}_0$. Let $N=\max(n,r)$. Then:

$$m/10^n + p/10^r = (m \cdot 10^{N-n} + p \cdot 10^{N-r})/10^N$$

2. The numerator is an integer and N is a non-negative integer, so the sum is in D.
3. The product is:

$$(m/10^n)(p/10^r) = mp/10^{n+r}$$

4. Again the numerator is an integer and the exponent is non-negative; therefore the product is in D.

Self-Study 3.8: Finite decimals and computer arithmetic

8b · A prime factor missing from powers of 10

1. If 1/3 were in D, then $1/3=m/10^n$ for some integer m and $n \geq 0$. Cross-multiplying gives $10^n=3m$.
2. The right side is divisible by 3. But 10^n leaves remainder 1 when divided by 3: for $n \geq 1$, expand $10^n=(1+9)^n$; every term after the initial 1 is divisible by 3. For $n=0$ the value is also 1. Contradiction.
3. Thus **1/3 is not a finite decimal fraction**.

Self-Study 3.8: Finite decimals and computer arithmetic

8c · Bound the discarded tail

1. Let m be the greatest integer not exceeding $10^n x$. By definition, $m \leq 10^n x < m+1$.
2. Divide through by $10^n > 0$:

$$m/10^n \leq x < m/10^n + 1/10^n$$

3. Since $x_n = m/10^n$, this says $0 \leq x - x_n < 10^{-n}$.
4. Also x_n has integer numerator m and denominator 10^n, so $x_n \in D$.

Self-Study 3.8: Finite decimals and computer arithmetic

8d · Rounding breaks associativity

1. For $a = 10^{16}$, $b = -10^{16}$, and $c = 1$, the first grouping cancels the large values exactly before adding 1:

$$(a + b) + c = 0 + 1 = 1$$

2. Near 10^{16}, adjacent double-precision values are 2 apart. The exact value $b+c = -(10^{16}-1)$ lies halfway between representable values and rounds to -10^{16} under round-to-nearest, ties-to-even.
3. Therefore:

$$a + (b + c) = 10^{16} + (-10^{16}) = 0$$

4. The results differ, so computer floating-point addition is **not associative** in general. The real-number associative law remains true; rounding after intermediate operations is what changes the computed result.

Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9a · Iterate the recurrence

1. Start with $F_1 = 1$ and $F_2 = 1$.
2. Then $F_3 = 1+1 = 2$, $F_4 = 2+1 = 3$, $F_5 = 3+2 = 5$, and $F_6 = 5+3 = 8$.
3. Continue: $F_7 = 13$, $F_8 = 21$, $F_9 = 34$, $F_{10} = 55$.
4. **Answer:** 1, 1, 2, 3, 5, 8, 13, 21, 34, 55.

Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9b · Read the diagonal sums

1. The first eight rows of Pascal's triangle are:

Row n	Coefficients C(n,k), k=0,...,n
0	1
1	1, 1
2	1, 2, 1
3	1, 3, 3, 1
4	1, 4, 6, 4, 1
5	1, 5, 10, 10, 5, 1
6	1, 6, 15, 20, 15, 6, 1
7	1, 7, 21, 35, 35, 21, 7, 1

2. Compute the specified sums (terms with k>n−k are zero):

$$D_0 = 1;\ D_1 = 1;\ D_2 = 1 + 1 = 2;\ D_3 = 1 + 2 = 3$$

$$D_4 = 1 + 3 + 1 = 5;\ D_5 = 1 + 4 + 3 = 8;\ D_6 = 1 + 5 + 6 + 1 = 13;\ D_7 = 1 + 6 + 10 + 4 = 21$$

3. These are the Fibonacci numbers F_1 through F_8: D_n appears to equal F_{n+1}.

Self-Study 3.9: Fibonacci numbers in Pascal's triangle

9c · Match the Fibonacci recurrence

1. **Base cases:** D_0=C(0,0)=1=F_1, and D_1=C(1,0)+C(0,1)=1+0=1=F_2. Two are needed because both recurrences depend on the previous two terms.
2. For the recurrence, use Pascal's identity inside the sum:

$$D_{n+2} = \sum_{k=0}^{n+1} C(n+2-k,k) = \sum_{k=0}^{n+1} [C(n+1-k,k) + C(n+1-k,k-1)]$$

3. The first sum is D_{n+1} (out-of-range end terms are zero). In the second, set j=k−1; its non-zero terms become C(n−j,j), summing to D_n. Hence D_{n+2}=D_{n+1}+D_n.
4. **Induction step:** assume D_n=F_{n+1} and D_{n+1}=F_{n+2}. Then D_{n+2}=F_{n+2}+F_{n+1}=F_{n+3} by the Fibonacci recurrence.
5. The two base cases and this step prove D_n=F_{n+1} for every n∈$\mathbb{N}_0$.

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